
When Do Language Models Check the False Steps in Their Own Reasoning?

Heterogeneous Boundaries and a Universal Recompute Wall

Anonymous / [TODO: author list]
[TODO: affiliations]

Abstract

When a language model continues a reasoning trace that already contains a false step, does it catch the error or quietly build on it? We study this with an audited perturb-and-validate instrument: a statement is planted into a model’s own proof trace on formally checkable logic problems, audited for whether it is true or false under the closed world, and the model’s unprompted continuation is formally validated.

Responses to a planted false step are strongly model-dependent. On the primary model, Qwen2.5-7B, explicit rejection is a visibility event: a planted falsehood is rejected in roughly 0.25 of continuations when its contradiction is a directly readable premise and in about 0.03 one hop away, flat beyond, and genuine multi-hop checking exists but is rare and ungraded by proximity. This visible-contradiction rejection pattern does not transfer: on three of four other open-weights models it significantly reverses, with rejection *lower* when the contradiction is visible than when it is a hop away.

Among the models that absorb planted falsehoods, evidence distance does not predict absorption but derivational usefulness does: a usable falsehood is reused near 0.8 regardless of how close its refutation sits, an inert one near zero. Absorption itself declines to near zero with scale within one family (0.81 at 7B to 0.01 at 32B in Qwen2.5) and varies by training recipe across labs at a matched harness.

Two results hold across scale and lineage. No tested model re-verifies an already-computed value: recomputable falsehoods are absorbed near 1.00 even by frontier models that reject re-readable ones. And verbal rejection is a weak proxy for checking: a frequency-matched non-refuting rule elicits the same rejection rate as genuine refutation, yet those rejections fabricate a rule absent from the world, so only a derivation-licensed split separates checking from lexical cueing.

Single-model conclusions about how reasoning errors are handled do not transfer across model, scale, or training recipe; computed intermediate values are the one universal blind spot. We frame the in-stream-versus-fresh-turn part of this last boundary as corroboration of concurrent work on the self-correction blind spot, not as a discovery, and our controlled tests run on a single model family and a synthetic logic domain.

1 Introduction

A model that continues its own flawed reasoning can silently absorb the flaw. Language models generate long reasoning traces one token at a time, and a false intermediate step, whether hallucinated, mis-copied, or adversarially planted, does not announce itself. The question this paper asks is behavioral and narrow: when a false statement already sits in a model’s own trace and the model keeps going with no external prompt to check, does it reject the statement or reuse it. The answer depends on the model. On Qwen2.5-7B-Instruct, the response is governed by two things, whether the contradiction is directly visible in the trace and whether the false step is useful for finishing the proof, and not by how “checkable” the

error is in principle. Across four further open-weights models this structure largely does not transfer: the visibility pattern reverses on three of them, and absorption itself disappears at 32B scale. Two regularities hold in every model we tested: none re-verifies a value that must be recomputed, and verbal rejection is a poor guide to genuine checking.

Evidence distance does not predict the response. A natural hypothesis is that rejection is graded by inferential distance to the refuting evidence. A matched-distance test on Qwen2.5-7B-Instruct, laid out as six distance-linked predictions, gives no support for it: five fail, and the point estimates for the two that carry the gradient story run opposite to it. The one prediction the test supports is not a distance gradient but a polarity control, that affirmative and negated false statements are absorbed equivalently. What predicts the response instead is whether the contradiction is directly visible and whether the false step is useful for finishing the proof. The mechanics of how the predictions were fixed in advance are collected in Appendix A.

Visibility and usability boundaries govern the anchor model. On the anchor model, explicit rejection of a planted falsehood is abundant when the contradiction is a directly readable premise, roughly 0.25 of continuations at measured distance $d=0$, and collapses about eightfold to 0.03 one hop away, staying flat for larger distances. Separately, silent downstream reuse of a planted falsehood tracks whether that falsehood is derivationally needed to finish the current proof. A usable falsehood is reused near 0.8 whether its refutation is one hop or infinitely many hops away, and an inert one is reused near zero. Both are boundary conditions on when in-stream self-checking happens at all, and neither is a graded distance gradient.

The boundary conditions do not replicate across models. The single-model-family base is the obvious weakness, so we ran a multi-model replication that repeats the three boundary conditions on four more open-weights models: Llama-3.1-8B, OLMo-2-7B, Qwen2.5-1.5B, and Qwen2.5-32B, with its predictions again fixed before the data were generated and its verdict independently re-checked from the raw data. Zero of the four fully replicate. The visibility cliff is the sharpest casualty: on three of the four models the effect significantly reverses, so explicit rejection is *lower* when the contradiction is directly visible than when it is a hop away, the opposite of the anchor. The usability asymmetry is the most portable of the three, holding on two of the four, but it is scale-bounded: usable-error absorption falls from 0.71 at 1.5B to 0.81 at 7B to 0.01 at 32B, so the effect essentially vanishes with capability within a single model family. Because the same pipeline reproduces the Qwen2.5-7B anchor exactly, this is a property of the models, not of the analysis. We therefore present the single-model-family limitation not as an open caveat but as a positive finding: how a model handles a planted falsehood in its own trace is model-dependent and scale-dependent.

Two results do survive across scale and lineage. First, a recompute boundary. In a program-execution regime, no model we tested, including two frontier models that reject re-readable planted errors near totally, re-verifies a value it must recompute; recomputable falsehoods are absorbed at 0.98 to 1.00 everywhere. Second, an instrument caution that holds wherever we can check it. Verbal “rejection” cannot by itself distinguish a genuine check from lexical cueing: a frequency-matched but non-refuting rule elicits the same rejection rate as genuine refuting evidence, and every one of those frequency-matched rejections fabricates a rule that is absent from the world. Only a derivation-licensed split, which asks whether the model’s own text contains a valid derivation, separates the two.

Scope guard. All of these findings concern non-deliberative, in-stream continuation on synthetic first-order-logic worlds. We do not claim they extend to explicit deliberation or thinking modes, to natural prose, or to tasks outside the audited-logic substrate; models with a separate deliberation channel are deliberately excluded from the controlled tests. We also frame the in-stream-versus-fresh-turn part of the production boundary as corroboration of concurrent work on the self-correction blind spot, discussed in Section 9, not as a discovery of ours.

Contributions. We contribute (i) an audited perturb-and-validate instrument that truth-audits every plant, formally validates every continuation, and measures refutation distance mechanically rather than by design label (Section 2); (ii) evidence that inferential distance does not predict in-stream rejection or reuse (Section 3); (iii) two boundary conditions, visibility and usability, on the anchor model (Sections 4, 5),

together with a multi-model result that they largely do not replicate; (iv) a recompute boundary that does survive across scale and lineage (Section 6); and (v) an instrument caution that verbal rejection is lexically cued (Section 7). We read the whole as a boundary-condition map with an honest account of how far each boundary travels.

2 Protocol: audited perturb-and-validate

We plant a statement into the model’s own trace and formally validate what it does next. The unit of analysis is a single statement inserted into a model’s in-stream reasoning trace on a formally checkable first-order-logic task in the PrOntoQA family (Saparov & He, 2023; Saparov et al., 2023), after which the model continues with no external correction. Two audits bracket every row. Before generation, each planted statement is truth-audited against rule closure in the closed world, so we know whether the plant is actually true or false rather than assuming it. After generation, a formal validator assigns each continuation to one of a set of mutually exclusive, precedence-ordered outcome classes. The classes, in precedence order, are generation-failed, unparsed, derailed (the final answer differs from the target), injection-dependent (at least one step is derivable only via the planted atom, but the final answer still matches the target), parroted (the final answer matches despite an underivable step), and closure-valid completion (every step is entailed by the true premises and prior valid steps under rule closure). Two facts anchor these definitions for the rest of the paper. Closure-valid is a closure-based standard, so a single continuation sentence may legally jump several hops; it is not a strict replay. And the word “recovery” is reserved throughout for one defined metric, strict repair, introduced below; we otherwise say the model rejected, absorbed, flagged, or reused a plant.

The truth audit is load-bearing, not decorative. An earlier version of this instrument treated a “wrong-category” perturbation family as a family of falsehoods. Under the audit, 355 of its 390 members are actually true and only 35 are false, so a headline built on it would have measured behavior on mostly-true statements. On the matched-distance grid used here the audit is clean: every planted-false cell is audited false and every anchor cell audited true. We report the 355-of-390 inversion in the protocol, not as a result about models, because it is the reason the audit exists.

The design matrix crosses the factors so they are pinned, not confounded. The matched-distance grid crosses truth status, polarity, derivational usability, and measured inferential distance across 24 cells, with parse gaps at most 0.67 percentage points across matched cells, so visibility, usability, and polarity can be read separately rather than being tangled together (Table 1). This is the visual answer to the reviewer question of whether the effects are confounded: each row of Table 1 holds the other factors fixed.

Distance is measured per row, not assumed from the design label. We do not trust the designer’s intended distance. A mechanical breadth-first search over rule applications computes, for each row, the shortest number of hops from the visible-prefix state to the evidence that would refute the plant. On the generated worlds used for the controlled results, designed and measured distance agree and the global design point is clean, with zero of 450 rows carrying a finite refutation distance. The audit also exposes why a legacy distance ladder cannot support a dose-response reading: its designed one-hop condition measures at distance zero on 14.9 percent of rows and at distance two or more on 8.4 percent, and its multi-hop rungs are heavily smeared, so any gradient over the designed label there is an artifact and we re-bin by measured distance. Measured distance is rule-BFS distance in the code-native fragment, not hop count in the model’s own written proof; we keep that distinction explicit.

One box fixes every dependent-variable definition and the backend rule. To keep the instrument honest we fix, in one place, the definitions of the stated-complement rejection measure, closure-valid and hop-sound validity, the strict-repair rate, and derivational injection-dependence, together with two house rules that bind every number in the paper. The first rule is that generation never mixes inference backends within a comparison, because the verbalized-doubt measure is backend-sensitive by 8 to 15 percentage points; the second is that doubt is reported only as a contrast, never as an absolute rate, for the same reason. The verbalized-doubt measure is a lexical proxy that we demoted to descriptive before review, and Section 7 shows why.

Table 1: Design matrix (schematic). The matched-distance grid crosses truth status, polarity, derivational usability, and *measured* refutation distance, so each factor can be read with the others held fixed. Full per-cell rates are in Appendix E. **[TODO: populate the schematic with the exact 24-cell layout from the matched-distance grid report before submission.]**

Factor	Levels	Held fixed by	Measured?
Truth status	audited-true / audited-false	closed-world closure audit	yes
Polarity	affirmative / negated	matched pairs	yes
Usability	usable / inert	derivational need for target	yes
Distance d	0, 1, 2, 3, 5, ∞	rule-BFS per row	yes (not designed)

Planting is justified because the modal natural error lives exactly where in-stream checking is hardest. A design that plants errors is only interesting if it plants the errors models actually make. We audited every unperturbed rollout in a corpus of 3,715 gold rollouts and found 236 natural invalid entity-facts. The planted families cover 93 percent of the natural invalid entity-facts, and the single modal natural error, at 85 percent, is an affirmative-unentailed category whose refutation is more than eight hops away, which is exactly the global design point; locally refutable natural errors are rare, at 8 percent. The natural distribution also reproduces the paper’s central asymmetry, with naturally usable errors consumed downstream about eleven times more often than inert ones (Section 5). One honest gap travels with this: the single most frequent natural error *object* is a fabricated or mis-spliced rule rather than a false entity-fact, which no planted family emulates except in inert form, so we scope the paper’s claims to entity-fact falsehoods.

The grid is auditable end-to-end and fails closed. The matched-distance grid generated 2,880 worlds across 360 families with zero world-audit failures and 11,577 validated rows, all with unique run identifiers. Injection audits reject fail-closed rather than silently passing a mis-audited plant, including 57 rows dropped for a measured-distance mismatch. The multi-model replication pipeline re-audits the same way, with zero world-audit failures on every model (Section 5).

3 Evidence distance does not predict responses to planted errors

A distance gradient does not predict rejection or reuse. A natural hypothesis is that errors get caught when the disproving evidence is nearby: rejection and reuse should both be graded by how far the refuting evidence sits from the plant. A matched-distance test on Qwen2.5-7B, holding everything but distance fixed across six distance-linked predictions, finds no support for it. Rejection is flat to slightly rising across distance rather than falling (slope +0.011, $p=0.79$), and reuse is highest one hop away rather than farthest (slope -0.248 , $p=0.996$); both run opposite to a gradient. The one distance-linked prediction the test supports is not a gradient but a polarity control: affirmative and negated false plants are absorbed equivalently, within a ± 0.10 margin (diff +0.054, $p=0.0012$).

The polarity equivalence holds under the stricter validator too. A reader might worry that this surviving result rests on the permissive closure-valid standard we elsewhere demote. It does not. Under the strict hop-sound standard the polarity equivalence still holds within the ± 0.10 margin, with a difference of -0.042 and a TOST $p=8 \times 10^{-5}$; the point sign flips but the equivalence conclusion is unchanged. We state this as “equivalent within ± 0.10 ”, never as “identical”, because the equivalence fails at the tighter ± 0.05 margin and the base rates are low.

Two downstream contrasts carry different evidential weight. Two contrasts the rest of the paper builds on differ in how directly the matched-distance test speaks to them, and we mark the difference rather than treating them alike. The $d=0$ visibility contrast of Section 4 is measured directly but was left direction-open, so it motivates the visibility picture rather than clinching it. The reversed reuse trend of Section 5, where reuse is highest one hop away, runs opposite to the gradient prediction and does not reach

a confirmatory threshold under matched construction, so we report it as a suggestive observation and never describe it as a confirmed effect.

No strict metric reverses a conclusion the paper relies on. We scanned the contrasts the paper leans on under the strict hop-sound standard and found no relied-on reversal; the one sign flip, in the polarity equivalence, strengthens rather than weakens the case, because it moves the estimate toward zero difference. The strict tables are in Appendix E.

4 Boundary condition one: visibility (on the anchor model)

On Qwen2.5-7B, rejection is a step function of visibility. When the statement that would refute a planted falsehood is a directly readable premise, at measured distance zero, the anchor model rejects the plant in about 0.25 of continuations; one hop away the rejection rate collapses roughly eightfold to 0.03, and it is flat for larger distances (Figure 1). Crucially, closure-valid completion is flat across distance, with a $d=0$ versus $d=1$ difference of -0.002 that is not significant, so the cliff is a change in whether the model *objects*, not a change in whether it can finish the problem. This contrast is measured directly but was left direction-open, so it motivates the picture rather than clinching it.

The $d=0$ signal is pure premise read-out, not proof search. The abundant rejection at distance zero is entirely the model reading back a contradiction that is already written in front of it. Within the licensed split, which asks whether the model’s own text contains a valid derivation, all 111 of 111 $d=0$ rejections are premise read-out with zero written derivations. This is why we call the effect visibility rather than checking: at $d=0$ the model sees the contradiction, it does not search for it.

Genuine multi-hop checking exists, but it is rare and not graded by proximity. We deliberately soften the strongest version of this claim. It is not true that the model never performs even one hop of proof search: the licensed split finds 235 complete written refuting derivations at designed distance one or more, 95 of which require at least two real hops, and among rejections whose complement is genuinely derivable the licensed fraction is about 0.79. What is true is that this genuine checking is rare within any distance-one-or-greater cell, a few percent, and it is not graded by proximity: the licensed-derivation rate is flat across distance with no peak near the plant. So the correct statement is that genuine distance-checking exists but is rare and non-graded, and the large $d=0$ effect is visibility, not proof search.

The one place genuine checking spikes is a cheap, useless hop. The exception is instructive. When a plant is inert, useless for the proof, and its refutation is exactly one cheap hop away, the model does check: that cell shows a rejection rate of about 0.19 that is 97 percent licensed genuine one-hop checking, and the checking decays with distance to near zero by three hops. This is a suggestive observation whose natural reading is that the model checks when checking is cheap and the error is not useful to it. That reading does not hold up: the usefulness of a planted statement does not detectably change genuine checking. In a matched-pair construction that pits a useful against a useless version of the same planted statement on matched worlds (the question-flip experiment), the checking gap is null (gap $+0.004$, 95% CI $[-0.002, +0.010]$), and the earlier exploratory one-hop spike does not replicate under this matched construction. The forward-looking, causal version of this “does the model’s goal change whether it checks” question is a separate line of work (Section 9), not a claim we make here.

A second, model-graded dependent variable reproduces the cliff. To rule out a regex artifact we re-scored all 11,577 rows with a 32B judge model. The judge reproduces the cliff, with judge-flagged doubt about 0.13 at $d=0$ versus 0.03 at $d=1$, a factor of about four, then flat, and it agrees with the lexical measure on 98 percent of rows. The judge’s separate explicit-rejection measure carries a weak, gradual tail beyond one hop, which is consistent with the rare non-graded checking above rather than with a graded distance law; we report the doubt measure as the clean cliff and note the tail.

The nearest rival account is addressability, and we stake the claim where it is neutralized. A concurrent account attributes such effects to *addressability*, the presence of a discrete referent to act on,

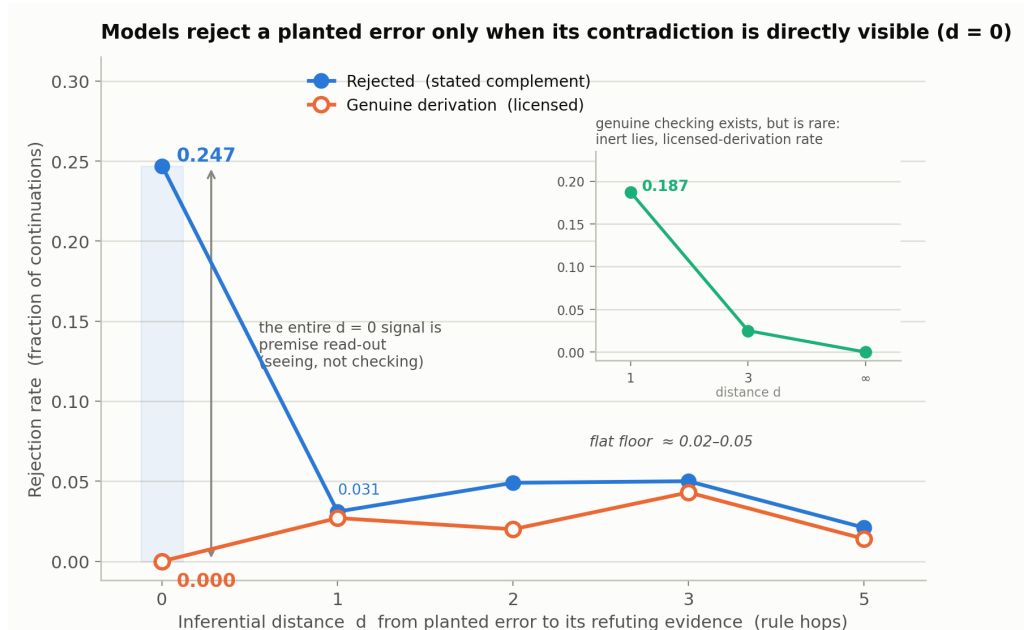


Figure 1: **The visibility cliff on Qwen2.5-7B, and why it is “seeing” not “checking”.** Explicit rejection (stated complement) of a planted falsehood peaks at measured distance $d=0$ (0.25) and collapses to a flat floor for $d \geq 1$ (0.03 at $d=1$). The licensed-only overlay, rejections backed by a genuine written derivation, is *zero* at $d=0$: the entire $d=0$ signal is premise read-out. Genuine multi-hop checking is not absent (235 written derivations exist at $d \geq 1$, 95 needing two or more hops) but it is rare and not graded by proximity. The inset shows the one real distance-check, the inert one-hop cell (0.19 rejection, 97% licensed), decaying with distance. *This figure is Qwen2.5-7B only; [TODO: add a per-model strip from the multi-model replication showing the cliff reverses on Llama-3.1-8B / Qwen2.5-1.5B / Qwen2.5-32B.]* Licensed numbers are the strict v2 classifier.

which coincides with visibility exactly at $d=0$ (Chen et al., 2026). We concede the coincidence at $d=0$ and therefore do not rest the claim there; we stake it on the flatness for $d \geq 1$, where a discrete refuting referent still exists in the world but rejection has already collapsed.

This visibility cliff is specific to Qwen2.5-7B. Everything in this section is measured on the anchor model, and the multi-model replication of Section 5 shows the cliff does not generalize: on three of four further models the $d=0$ contrast significantly reverses. We keep the visibility law scoped to Qwen2.5-7B and treat the reversal itself as a finding.

5 Boundary condition two: usability, and what replicates

Absorption does not fall as the refutation gets closer. We lead with the flatness rather than the usable-versus-inert gap, because the flatness is what kills the tautology that the model reuses errors it cannot refute. On Qwen2.5-7B, derivational injection-dependence for a usable plant, the rate at which the model silently reuses it downstream, is 0.86, 0.77, and 0.79 at one, three, and infinitely many hops, pooling to about 0.81. The refutation being adjacent does not help. The prediction that reuse rises with distance failed outright, and the trend we did see runs the opposite way, with reuse highest at one hop, exactly where the model could most cheaply refute the plant.

Inert falsehoods are essentially never reused, and the strict standard sharpens the split. Where a usable plant is reused near 0.8, an inert one is reused near zero, below 0.03. The dichotomy survives the strict hop-sound validator: usable plants almost never produce a hop-sound derivation, between 0.00 and

0.01, which is the point, they are reused rather than re-derived, while inert plants sit higher. So usability, not distance, is the axis that governs silent reuse on the anchor model.

The asymmetry is not an artifact of planting: it appears in the model’s own natural errors. In the audit of unperturbed rollouts, natural errors that happen to be usable are consumed downstream about 0.54 of the time versus about 0.05 for inert ones, roughly an elevenfold gap in the same direction as the planted 0.81 versus at most 0.02. The natural rate is lower than the planted rate and the natural cell is small and not position- or length-matched, so we read this as reproducing the asymmetry qualitatively, not as an equality of rates. The 32B judge shows the same asymmetry, flagging inert plants with doubt about nine to fourteen times more often than usable ones.

The boundary conditions do not replicate across models. The single-model-family base is the weakest part of the story, so we run the three boundary conditions on four further open-weights models, with the pass criteria fixed in advance: the $d=0$ visibility cliff, the usability gap, and flatness in distance. The rule set in advance for the overall claim was that at least three of the four models must fully replicate all three conditions. Zero of four do (Table 2). The result was adjudicated with the same locked machinery, which reproduces the Qwen2.5-7B anchor exactly, and then independently re-verified from raw data with every number reproducing to three decimals, so the non-replication is a property of the models rather than the analysis.

The visibility cliff reverses, and flatness fails everywhere. On the visibility cliff, three of the four models show a statistically significant reversal of the anchor effect: explicit rejection is lower when the contradiction is directly visible than when it is a hop away, by -0.14 on Llama-3.1-8B, -0.14 on Qwen2.5-1.5B, and -0.07 on Qwen2.5-32B, position-consistently. Flatness fails on all four, driven by the same fact: stated-complement is not flat in distance in the new models, it is elevated one hop away and declines, which is the “checking model” reading and the opposite of the anchor’s flatness.

Usable-error absorption is real but scale-bounded: it collapses by 32B. The usability gap is the most portable of the three, passing on Llama-3.1-8B ($+0.92$) and Qwen2.5-1.5B ($+0.70$). But it is bounded by scale. Within the Qwen family, usable-plant absorption falls from 0.71 at 1.5B to 0.81 at 7B to 0.01 at 32B, and the 32B model, rather than absorbing a usable false premise, tends to re-derive (or state) the true value or express doubt. The gap at 32B is significant but about twenty-three times below the $+0.30$ magnitude threshold we set in advance, so we read it as a genuine null: within the Qwen family, absorption via derivational usability appears to be a small-to-mid-scale phenomenon that vanishes with capability. OLMo-2-7B could not sustain the base task, with a greedy gold-solve rate of 0.20 that left its grid at 13.5 percent of target, so its slots are underpowered and reported but not scored; the program verdict is identical whether it is scored as a failure or excluded.

We read the non-replication as the paper’s central scope finding. How a model handles a planted falsehood in its own trace is model- and scale-dependent. The visibility law and the flatness framing are Qwen2.5-7B-specific; the usability asymmetry is the most portable but is scale-bounded and not universal. This converts the single-model-family limitation from an open caveat into a positive result about where these behaviors do and do not hold.

6 Boundary condition three: production and recompute

Absorption is a property of in-stream continuation, and this part corroborates prior work. The same audited-false plant that Qwen2.5-7B absorbs at 0.86 while continuing its own stream is absorbed at about 0.15 when the identical content is re-presented as a fresh chat turn, and at about 0.00 once restatement of the plant is suppressed; closure-valid completion rises correspondingly as the model re-plans through the true chain. A diagnostic shows the two failure modes trade off, so this is not fixable by prompt tightening. We frame this in-stream-versus-fresh-turn boundary as corroboration of concurrent work on the self-correction blind spot (Tsui, 2025; Chen et al., 2026), not as a discovery. Our delta over that work is

Table 2: **Multi-model replication: zero of four fully replicate.** Per-model verdicts for the three boundary conditions. Visibility = the $d=0$ cliff (passes iff significant and point diff $\geq +0.10$); Usability = the usable–inert gap (passes iff significant and diff $\geq +0.30$); Flatness = flatness in distance (conjunctive equivalence test). The anchor (Qwen2.5-7B, top row) is the reference, not re-tested here. Bracketed values are cluster-bootstrap 95% CIs. Pass criteria and thresholds were fixed in advance; see Appendix G for integrity figures.

Model	Visibility ($d0-d1$)	Usability (usable–inert)	Flatness	Fully replicates?
Qwen2.5-7B (anchor)	+0.216 [+ .16, +.27] PASS	+0.799 [+ .76, +.83] PASS	pass	yes (ref.)
Llama-3.1-8B	−0.144 [−.19, −.10] REV.	+0.923 [+ .90, +.95] PASS	fail	no (1/3)
Qwen2.5-1.5B	−0.142 [−.22, −.07] REV.	+0.700 [+ .66, +.74] PASS	fail	no (1/3)
Qwen2.5-32B	−0.069 [−.10, −.04] REV.	+0.013 (collapses)	fail	no (0/3)
OLMo-2-7B	+0.192 INVALID	+0.442 INVALID	invalid	no (underpowered)

Per-condition: Visibility 0/4, Usability 2/4, Flatness 0/4; no condition reaches the $\geq 3/4$ bar. *Overall:* 0/4.

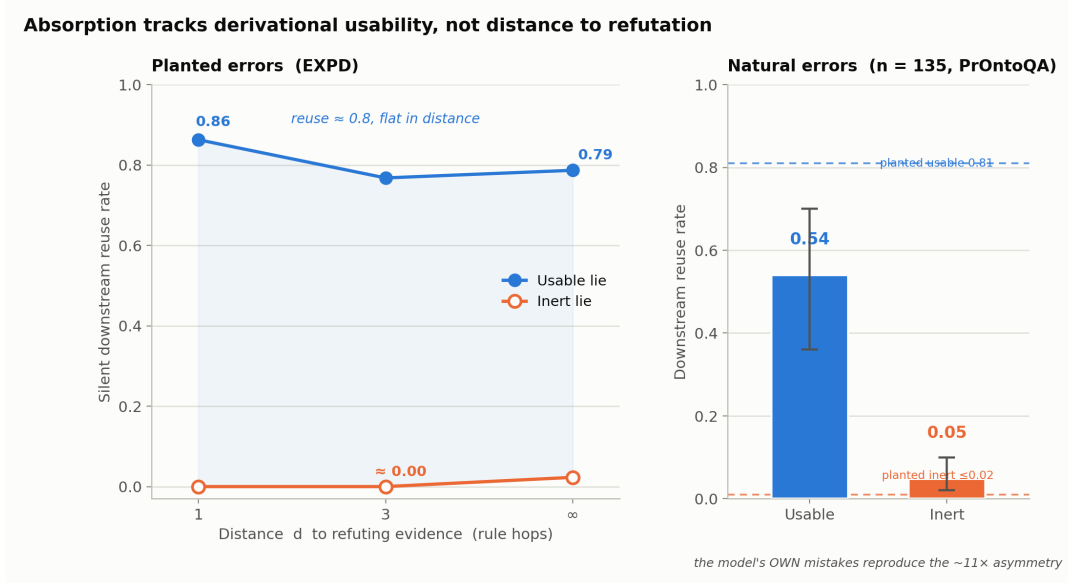


Figure 2: **Usability, not distance, governs silent reuse on Qwen2.5-7B.** Left: silent downstream reuse of a *usable* planted falsehood is about 0.8 and does not decay as the refuting evidence moves closer (0.86 / 0.77 / 0.79 at $d=1/3/\infty$); *inert* plants sit near zero. Right: the same asymmetry appears in the model’s own natural errors (usable 0.54, inert 0.05), reproducing the planted direction. The usable trend is a suggestive observation, not a confirmed effect (the gradient prediction runs the other way). Natural cell is small and unmatched, so this is a qualitative match. *Qwen2.5-7B only*; **[TODO: overlay the per-model replication showing the usability gap passes on Llama-3.1-8B / Qwen2.5-1.5B but collapses to 0.013 at 32B (Table 2).]**

a silent-reuse measure that answer-naming judges miss, a near-total magnitude, 0.86 to about zero rather than a few points of dilution, and an audited-logic substrate.

Re-read rejection is training-recipe-bound, a lab-level split rather than a capability ladder. Under an identical prefill harness, the current Anthropic chat lineage rejects every audited plant, with derivational injection-dependence of 0.000 in all twelve cells for each of two models, including the very cell where Qwen absorbs at 0.86, while the current OpenAI lineage still absorbs. In the one directly comparable format, instruct-mode on each model’s own traces with audited-false plants, echo-corrected reuse of a false category is 0.58 for gpt-4.1 and 0.01 for claude-sonnet-5, spanning the open-weights anchor of 0.35 in be-

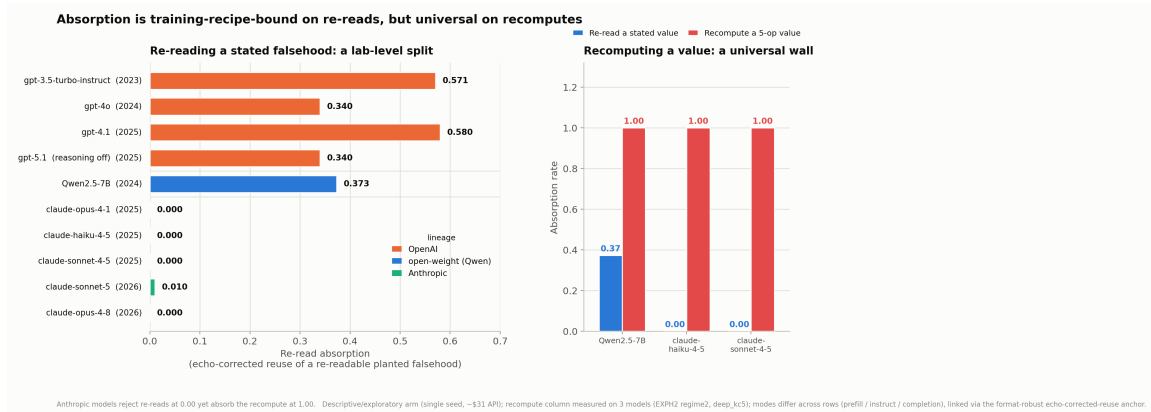


Figure 3: **Lineage split on re-reads; a universal wall on recomputes.** Left: echo-corrected reuse of a re-readable planted falsehood is lab-shaped and non-monotone in capability, with 2023 gpt-3.5-instruct (0.57) and gpt-4.1 (0.58) the deepest absorbers, the open anchor at 0.37 (the prefill-locked format value; the same Qwen comparator is 0.350 in the instruct format used for the cross-lab comparison in the body text), and every Anthropic model near 0.00. Right: on recomputes there is no split, the two Anthropic models that reject re-reads at 0.00 still absorb a five-operation recompute at 1.00. This whole panel is descriptive, single-seed, roughly \$31 of API; rows in the left panel use different modes (prefill / instruct / true completion) and are linked only through the format-robust echo-corrected-reuse anchor, so read it as a gradient, not one controlled ladder. Never read it as “Anthropic checks every time” (supported claim: rejects every audited falsehood, reuse 0.000) or “gpt-4.1 never checks” (only its reuse rate 0.58 is on record).

tween.[[PENDING: add Self-Attribution Bias 2603.04582 and Cross-Context Review 2603.12123 as stubs when this sentence is wired in Sec 9.](#)] The oldest generation point we could run under true text completion, 2023-era gpt-3.5-turbo-instruct, is the deepest absorber measured anywhere, at echo-corrected reuse 0.57, giving the lineage a non-monotone shape that complicates any clean training-recipe or monotone-compute story. We keep this arm descriptive and single-seed, and we scope “Anthropic best” to the re-read prefill regime, because those same models fail in a monitoring regime studied elsewhere.[[PENDING: cite Self-Attribution Bias 2603.04582 here once abs-verified \(it shows Anthropic models fail in the monitoring regime\).](#)]

But there is a universal recompute boundary: no model re-verifies a computed value. In a program-execution regime we can separate a falsehood whose true value can be *re-read* from an earlier line from one whose true value must be *re-computed*. The two frontier Anthropic models reject re-readable falsehoods near totally, at 0.00 to 0.22, yet absorb a five-operation recompute chain at 1.00, indistinguishable from Qwen; a one-operation recompute is already absorbed at 0.98 to 1.00. No model we tested re-verifies a value it must recompute. We distinguish a final-answer flip from re-derivation of the specific computed intermediate, so this is not contradicted by math self-correction results, and we scope the universality to the three models actually run on those cells until a broader run lands. This recompute boundary is the one result in the paper that holds across scale, family, and lineage.

A conjecture from the earlier draft is now confirmed. The earlier workshop draft conjectured that absorption is carried by continuation pressure rather than by the content of the plant. The bridge experiment above confirms it, as a continuity asset rather than a retraction: suppressing in-stream restatement removes absorption entirely.

7 Instrument finding: verbal rejection is a weak proxy

Raw rejection rate cannot tell a genuine check from lexical cueing. A frequency-matched but non-refuting lexical-overlap arm, which mentions the plant’s terms without actually contradicting it, elicits

stated-complement “rejection” at 0.12, statistically the same rate as genuine refuting evidence at 0.11, while an irrelevant template arm sits at 0.00. This lexical-priming confound was one we had hoped to rule out in advance; it was borne out instead, so the prediction that no such confound existed failed in the direction we had guarded against, which is exactly why raw rejection rate is not evidence that the model checked.

The two arms differ in kind, and only the licensed split reveals it. Same rate, different mechanism. When we ask whether the rejection is backed by a valid derivation in the model’s own text, the genuine refuting-evidence rejections are about 80 percent licensed, whereas every one of the 26 frequency-matched rejections is unlicensed and fabricates a rule that is absent from the world. So the correct statement is not that refuting-arm rejections are cued, most are not, but that raw rejection rate cannot separate a genuine check from a fabricated one, and only the licensed split can. This is the caution that discredits reading a face-value doubt or reflection classifier as evidence of checking (Figure 4).

The lexical proxy is unreliable in both directions. It over-fires, as above, and it also under-reads. On a malformed continuation tail from the certificate experiment, the lexical regex agrees with the 32B judge only 23 percent of the time, catching 21 of the 95 unparsed rows (where the judge reads 86 as rejection), against 98 percent agreement on the clean grid, so the regex both invents rejections that are fabrications and misses rejections that broke the parser. We therefore treat the 32B judge as the better dependent variable and doubt as descriptive only.

Under the strict standard, most closure-valid completions are not step-by-step derivations. When we require hop-soundness, that every asserted step be a single valid inference, the apparent correction rate on the false-attribute cells drops sharply: closure-valid rates of 0.90 to 0.96 fall to 0.07 to 0.25 hop-sound, so only about 11 to 27 percent of closure-valid completions are genuine step-by-step derivations. This strict pass covers 6,147 rows, verified three independent ways; an earlier log reporting 4,857 was a stale error and we do not use it.

Human validation of the validator is still pending. The validator’s labels are checked against a stored class on all rows, agreeing 1.000, but the planned human-blinded two-rater audit that would confirm the labels against people has not been run. We therefore hold the strict-repair credibility claim as pending human validation and mark it visibly rather than shipping it silently. **[PENDING E4 HUMAN VALIDATION: E4 blinded two-rater package (380–430 rows including 20 SRR positives), gating $\kappa \geq 0.8$ per axis; Table of human κ to be filled. Until then, strict-repair is reported as an instrument quantity, not a validated behavioral claim.]**

8 Stakes: the lineage panel, scoped

The lineage panel is descriptive and scoped, not a vendor leaderboard. The cross-lineage comparison of Section 6 uses one seed, is single-format only where it compares directly, and is anchored through the format-robust echo-corrected reuse metric. We present it as stakes, why in-stream absorption matters for chain-of-thought monitoring, with the universal recompute column as its non-partisan center, rather than as a ranking.

The load-bearing comparison is within one format. The single most defensible statement is the within-format one: same instruct format, each model’s own traces, audited-false plants, gpt-4.1 reuses a false category at 0.58 while claude-sonnet-5 reuses at 0.01. We never say the Anthropic models check every time; the supported claim is that they reject every audited falsehood in this harness, at closure-valid 0.97 to 1.00 and reuse 0.000, and the verbatim one-hop-check transcript is attributed to a single model. We never say gpt-4.1 never checks; only its reuse rate is on record.

The split is not memorization. The worlds use nonce predicates, and the same Anthropic models that reject re-readable plants absorb recomputable ones at 1.00, which a memorization account cannot produce.

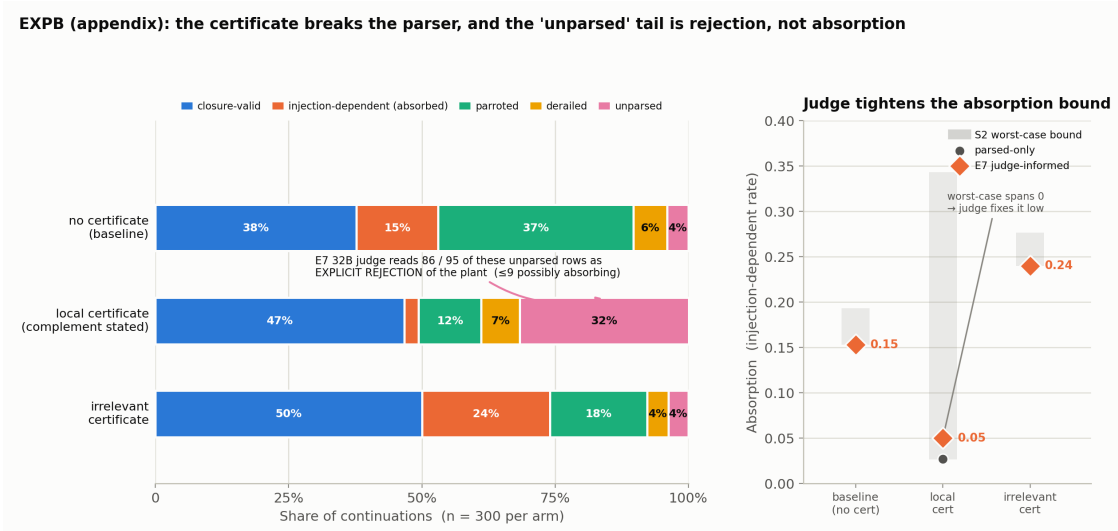


Figure 4: **Stating the contradiction in the prefix shifts continuation style rather than cleanly fixing the proof.** Left: adding a directly-contradicting local certificate drives the unparsed rate from 4.0% to 31.7%, draining the parsed buckets, rather than converting absorption into a clean corrective derivation. Right: the 32B judge reads 86 of 95 of those unparsed rows as *explicit rejection*, so the apparent absorption drop is real once adjudicated (about -0.10), but because the certificate is a directly readable contradiction this corroborates the $d=0$ visibility cliff of Figure 1 rather than a distinct counterevidence-accessibility effect. Single model (Qwen2.5-7B); doubt and strict-repair here are lexical-proxy contrasts, not evidence of a genuine check.

Table 3: **Evidence-tier map.** Each family of claims is labeled by how much weight it carries. Tier 2 now holds a verified non-replication, and tier 1 is scoped to Qwen2.5-7B where tier 2 tested it. Cumulative API spend across the descriptive arm was \$30.81.

Tier	Basis	Claims
1 anchor (controlled)	Qwen2.5-7B, predictions fixed in advance	Visibility cliff, usability asymmetry, lexical cueing. Scoped to Qwen2.5-7B per tier 2.
2 replication (controlled)	multi-model, predictions fixed in advance	Non-replication (verified): 0/4 fully replicate; visibility cliff reverses on 3/4; usability gap holds on 2/4 but collapses at 32B; flatness 0/4.
3 descriptive API	single-seed	Production boundary (corroboration), lab split, recompute boundary.
4 legacy / appendix	instrument lesson	Certificate experiment, legacy gap decomposition.

Every by-model claim carries an evidence tier. To keep the descriptive arm from being read as a controlled result, Table 3 labels each claim by how much weight it carries: controlled on the anchor model, controlled multi-model replication, descriptive single-seed API, or legacy appendix. The replication tier now carries the non-replication verdict, and the anchor-model tier is explicitly scoped to Qwen2.5-7B where the replication tested it.

9 Related work

Perturbing a model’s own reasoning stream is heavily precedented; our novelty is the audited package. Injecting or perturbing a model’s chain of thought and watching what follows is an established design (von Recum et al., 2026; Aravindan & Kejriwal, 2026; Shah et al., 2025; Amjith et al., 2025; Han

et al., 2026). The closest predecessors intervene on their own traces and report downstream accuracy or a doubt expression (von Recum et al., 2026), or measure faithfulness as stance consistency under output-level interventions (Han et al., 2026). We do not claim the inject-and-continue skeleton. What we add is the five-part audited package: a per-plant truth audit, a four-way formal validator, a measured rather than designed refutation distance, a derivation-licensed split, and a usability manipulation. Our lexically-cued-doubt result also cautions against reading a doubt-mediated “recovery” as genuine checking (von Recum et al., 2026).

Self-correction fails on a model’s own output, and we corroborate rather than discover that boundary. The self-correction literature establishes that models correct external errors better than their own (Huang et al., 2024; Tyen et al., 2024; Kamoi et al., 2024). Most directly, Self-Correction Bench names and quantifies the blind spot first, across many models with a controlled injection design (Tsui, 2025), and a chat-template-role account reproduces the same own-versus-external asymmetry (Chen et al., 2026). We concede priority on the boundary itself. Our production-boundary result corroborates and extends this to formally audited logic, adding a silent-reuse measure their answer-naming judges miss and a near-total magnitude. The prefill regime we probe is the in-stream continuation that autoregressive consistency sustains and that vanishes across a turn boundary (Qi et al., 2025; Lyu et al., 2026; Park & Kim, 2026).

Detection-without-action is known; our addition is the measured distance axis and the usability axis. Internal probes can detect a model’s own errors while behavior does not act on them, a diagnostic-not-causal dissociation (Yuan et al., 2026; Yee et al., 2024), with internal self-repair and correctness machinery documented separately (McGrath et al., 2023; Lad et al., 2024; Sun et al., 2026). Process and step verifiers exist as external modules (Lightman et al., 2024; Zheng et al., 2025; You et al., 2025); our question is instead whether the generator checks in-stream and unprompted. The nearest neighbor to our usability axis silently adopts simple hints while verbalizing complex ones (Chen et al., 2025)[[PENDING: home the simple-adopted / complex-verbalized dissociation on Unspoken Hints 2509.26041 once abs-verified; chen2025reasoning otherwise carries only the “under 20 percent verbalized” claim in P4](#)]; we instead manipulate graded derivational usability against a formal validator and measure silent downstream reuse, which we mark as the cleanest novel result and lead with its flatness across distance. Question-side premise-critique and conflicting-knowledge work (Li et al., 2025; Feng et al., 2026; 2025) contrasts with our trace-side plant.

Verbalized reasoning can be unfaithful, which grounds our lexical caution. Chain-of-thought text can misrepresent the true driver of an answer (Turpin et al., 2023; Lanham et al., 2023; Arcuschin et al., 2026), with under 20 percent of decisive hints verbalized (Chen et al., 2025), and invalid chains retain most of their gains (Schaeffer et al., 2023; Wang et al., 2023), which is why doubt-shaped text is not a check (Kambhampati et al., 2026). Against this lineage our contribution is a behavioral fabricated-rule signature specific to rejection text: essentially all control rejections invent a nonexistent rule, versus roughly 80 percent derivation-backed for genuine evidence, a citable caution that discredits face-value doubt classifiers[[PENDING: contrast against the prior probe-based rationalization-detection operationalization Catching Rationalization 2603.17199 once abs-verified](#)].

Computation-versus-validation, training recipe, and a held-out motivated-reasoning direction. A mechanistic analysis shows models compute arithmetic but fail to validate it, via surface consistency heads (Bertolazzi et al., 2025); we concede mechanistic priority and offer our universal recompute boundary as the behavioral corroboration[[PENDING: add Trust-but-Don’t-Verify 2606.05403 here as the behavioral corroborator of the recompute boundary once abs-verified](#)]. Our non-monotone cross-lineage pattern, where a mid-tier frontier model absorbs worse than the open anchor and the oldest model is deepest, complicates clean training-recipe accounts of the blind spot and the reflection-emerges- from-RL story (DeepSeek-AI et al., 2025; Yang et al., 2025; d’Aliberti & Horta Ribeiro, 2026)[[PENDING: the clean SFT\(bad\) / RL\(good\) spectral-radius account SPARC 2607.09803, which this non-monotone lineage complicates, TODO-verify by 2026-08-15](#)]. Finally, the usefulness of a planted statement does not detectably change genuine checking. A tempting account holds that checking is suppressed when the planted error is useful to the model’s current goal; its nearest analogue gates suppression on task compliance rather than derivational usefulness (Tsui, 2025)[[PENDING: cite must-add stubs Knowing-but-](#)

Not- Correcting 2605.05957, Trust-but-Don't-Verify 2606.05403, and choice- supportive-bias 2507.03120 once abs-verified]. A matched-pair construction that pits a useful against a useless version of the same planted statement on clean worlds (the question-flip experiment) shows no such effect: the checking gap is null (gap +0.004, 95% CI $[-0.002, +0.010]$), and the earlier exploratory one-hop contrast that suggested goal-sensitivity does not replicate under this matched construction. The forward-looking, causal version of the question, whether a model's motive can be made to change whether it checks, remains open, and we treat it as a separate line of work rather than a claim here.

10 Limitations

The controlled base is a single model family, and we now know the boundary conditions are model-specific. Every controlled verdict is on Qwen2.5-7B. The multi-model replication of Section 5 turns this from an open caveat into a finding: the visibility cliff and the flatness framing do not generalize, and the usability asymmetry, while the most portable, is scale-bounded. Readers should treat the anchor-model laws as scoped to that model unless and until further replication extends them.

The domain is synthetic logic, and the second regime is only piloted. The worlds are PrOntoQA-style first-order-logic problems, chosen for formal checkability, and the program-trace recompute regime is a pilot rather than a full experiment. A broader program-execution run would widen the recompute-universality claim beyond the models on those cells; we treat that run as optional for this venue and mark the scope accordingly.

The validator and judge carry caveats, and human validation is pending. The strict hop-sound standard prices in re-deriving the omitted gold step, which the true-plant control bounds; the global cell's near-zero strict-repair is partly structural, because the complement of a closed-world-only falsehood is itself underivable, so an explicit correction breaks closure validity; and verbalized doubt is a demoted lexical proxy reported only as a contrast. The planned human-blinded rater audit has not been run, so the strict-repair credibility clause is held pending. **[PENDING E4 HUMAN VALIDATION: E4 human κ audit gates the SRR credibility claim.]**

Perturbed problems over-represent long chains, and the generated worlds are easier than the natural cohort. Proof compression filters short chains out, so the perturbed set skews long, and the generated controlled worlds are easier than the natural cohort, with a manipulation anchor that does not pass its own equivalence gate by a small margin. The natural usable-versus-inert comparison is position- and length-unmatched and uses a small cell, so it is qualitative.

One error type is out of scope. The single most frequent natural error object is a fabricated or mis-spliced rule, not a false entity-fact, and no planted family emulates it except in inert form, so our claims are scoped to entity-fact falsehoods and the rule case is a stated gap.

11 Conclusion

A boundary-condition map, honestly scoped. Under an audited protocol, in-stream verification of a planted falsehood on one well-studied model is governed by visibility and usability rather than by any distance gradient. That map does not generalize: the visibility cliff is specific to Qwen2.5-7B and reverses elsewhere, and usable-error absorption collapses with scale. Two things survive. No model we tested re-verifies a value it must recompute, and verbal rejection is a weak proxy that a derivation-licensed split is needed to read. We offer the whole as a map of where in-stream self-checking happens, with an explicit account of how far each boundary travels, and we take the non-replication itself to be the most useful thing the audit produced.

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A Reproducibility and analysis provenance

This appendix collects the analysis provenance that the body reports in plain language: what was fixed in advance, the internal code names, the commit record, the correction structure, the amendment and deviation log, and the accounting of which results were confirmatory versus exploratory.

Registration mechanics. Predictions were fixed and committed before any rollouts were generated, on two occasions. (i) The anchor design registered the distance-gradient account on Qwen2.5-7B-Instruct as a Holm-corrected family of six hypotheses (plan PLAN2 v1.1, git commit `545b35db`, 2026-07-02T14:05:29-07:00, file `sha256 112da1a5...`). (ii) The multi-model replication registered the three boundary conditions and the “ ≥ 3 -of-4 models fully replicate” program rule, with per-condition pass thresholds (visibility diff $\geq +0.10$; usability diff $\geq +0.30$; flatness by conjunctive TOST), git commit `bdb098a1` (2026-07-24), adjudicated in `E1_ADJUDICATION.md` and independently re-verified in `E1_VERIFICATION.md`. A third registration governs the motive question-flip experiment (commit `3986290`, 2026-07-24T10:48:09-07:00), committed before generation.

Hypothesis registry (internal code $\rightarrow$ plain name in body). The body names none of these codes; the mapping is here for reproducibility.

Code	Plain name in body	Verdict
H1'	rejection falls with distance	failed (slope $+0.011$, $p=0.79$)
H4	reuse rises with distance	failed ($p=0.996$; reversed trend exploratory)
H2'	polarity control (aff. = neg.)	the one survivor ($p=0.0012$; TOST hop-sound 8×10^{-5})
H5'	no lexical-priming confound	failed in the guarded direction ($p=0.629$)
C-0	$d=0$ visibility contrast	registered but direction-open (motivating)
B1 / B2 / B3	visibility / usability / flatness	replication conditions (Table 2)

Experiment registry (internal code $\rightarrow$ descriptive name in body). EXPD = the matched-distance grid (anchor Qwen2.5-7B); E1 = the multi-model replication; E9 = the question-flip experiment (motive null); EXPB = the certificate experiment; EXPC = the polarity control; EXPE = the lexical-arm / dependent-variable registry; EXPH and EXPH2 = the cross-lineage and recompute API panels.

Correction structure. The anchor family used a Holm correction over its six tests; equivalence claims (polarity, flatness) use two one-sided tests; the multi-model replication uses family-cluster bootstrap confidence intervals; the question-flip primary uses a paired cluster bootstrap against a minimum effect size of 0.05.

Amendment and deviation log. Multi-model replication: no material deviations; OLMo-2-7B is reported but not scored (greedy gold-solve 0.199). Question-flip experiment: registration amendment `d367a97` folded the crossed verification factor into the confirmatory family and set the candidate pool to 750; implementation fix `f447bc5` extended an exhausted nonce-noun pool during CPU world construction before any outcome data existed (adjudicated benign, first 9,716 yields byte-identical); the confirmatory queue omitted the ITT sensitivity invocation, which the adjudicator ran separately (sign agrees, both fail the MES bar); the power curve is degenerate and plays no role in the verdict.

Confirmatory-versus-exploratory accounting. Fixed in advance and carrying full weight: the polarity control (the one surviving anchor test), and the three replication conditions with their thresholds. Fixed in advance but direction-open (motivating, not clinching): the $d=0$ visibility contrast, and the frequency-matched lexical arm. Exploratory, reported as suggestive only: the reuse-highest-at-one-hop reversal, and the cheap-one-hop checking spike, whose matched follow-up (the question-flip experiment) returns a null. Held pending external validation: the strict-repair credibility claim (pending the E4 human κ audit), and the question-flip null (branch (b), *provisional* pending the blind non-author classifier-transfer audit; if per-arm classifier error differs by > 0.05 that primary is reported classifier-confounded, not causal).

B Legacy gap decomposition (reviewer Q1)

The legacy one-hop-versus-global gap is not an artifact of $d=0$ mis-binning. Under identical measurement, excluding the 14.9 percent of rows that measure at distance zero keeps or widens every gap, because those rows are low-doubt and low-closure. The closure-valid gap moves from $+0.278$ to $+0.299$ once $d=0$ rows are excluded, and the strict hop-sound gap from $+0.087$ to $+0.095$. Identical measurement is not the same as matched claims: this removes distance mis-binning, not the polarity and lexical confound, which is answered by the polarity deconfound (Section 3) and the frequency-matched arm (Section 7) instead.

C The certificate experiment (reviewer W3)

Adding a directly-contradicting local certificate does not cleanly convert injection-dependent continuations into clean ones; it redistributes the whole continuation-style distribution into unparsed, from 4.0 percent to 31.7 percent. Under honest worst-case accounting of the unparsed rows, only the verbalized-doubt contrast survives, at a net $+0.37$ with a McNemar $p=7.70 \times 10^{-34}$ and a bootstrap CI of $[0.31, 0.43]$. The 32B judge then reads 86 of 95 of those unparsed rows as explicit rejection, which both tightens the worst-case bound to about -0.10 and re-attributes the effect to $d=0$ visibility, because the certificate states the complement in the prefix. The certificate experiment therefore corroborates the visibility cliff rather than a distinct counterevidence-accessibility claim.

D Polarity control (EXPC)

The affirmative-local cell carries the highest doubt of any cell, which rules out an account in which the negation token itself drives the apparent rejection.

E Strict metrics

The strict hop-sound standard roughly halves closure-valid completion everywhere and surfaces structure that the permissive standard hides. On the false-attribute cells, closure-valid rates of 0.90 to 0.96 fall to 0.07 to 0.25 hop-sound, so only 11 to 27 percent of closure-valid completions are genuine step-by-step derivations. The strict pass covers 6,147 rows, verified three independent ways. Table 4 gives the strict-versus-permissive accounting for the central cells. **[TODO: expand to the full per-cell hop-sound and SRR table from `s1_strict_hopsound_REPORT` before submission.]**

Table 4: Strict (hop-sound) versus permissive (closure-valid) on representative cells, Qwen2.5-7B. Source: `s1_strict_hopsound_REPORT.md` Sec 4, `STRICT_METRICS_REPORT.md`.

Quantity	permissive (closure)	strict (hop-sound)
one-hop falsehood, pooled valid	0.671	0.300
global falsehood, pooled valid	0.393	0.213
aff_false rejection @ $d=0$	(closure flat)	0.247
usable reuse @ $d=1$	0.863	≈ 0.000

F API panel detail

The descriptive API arm (Section 6, Section 8) spans EXPH and EXPH2, with a cumulative spend of \$30.81 against a \$100 hard stop, full per-cell rates, and exclusion records for models no longer served.

G Multi-model replication detail

Table 2 summarizes the pre-registered replication; Table 5 gives its integrity figures. Every model passed its world audits with zero failures; OLMo-2-7B could not sustain the base task and is underpowered. The adjudication was independently re-verified from raw data, with every headline number reproducing to three decimals.

Table 5: Replication integrity. Source: `E1_ADJUDICATION.md` Sec 2, `E1_VERIFICATION.md` Sec 4.

Model	oversample	gold solve	eligible rate	grid rows	world-audit fails
Qwen2.5-1.5B	4×	0.883	0.283	6,477	0
Llama-3.1-8B	2×	0.982	0.541	6,600	0
OLMo-2-7B	4×	0.199	0.044	893	0
Qwen2.5-32B	2×	1.000	0.995	6,600	0

H Response-to-reviewer map and dependent-variable definitions

A mapping from the earlier reviewer concerns (W1–W7, Q1–Q4) to the experiments that address them, together with the full dependent-variable definitions and the 18-item disclosure checklist, is reproduced from the revision plan and the validator plan. **[TODO: paste the W1–W7/Q1–Q4 crosswalk and the validatorPlan Part 2 checklist here before submission.]**